

LAB 10

Study of IP Addressing and Sub-netting.

1. Find default subnet masks, network bits, hosts per subnet, no of subnets, subnet number, 1st Valid IP Address, Last Valid IP Address, and broadcast address.

ci). 8.1.4.5/16
 Class : A (1-126)
 Default mask (Class A) : 255.0.0.0
 Given mask : /16 : 255.255.0.0
 Network bits = 16, Host bits = 16
 Hosts per subnet = $2^{16} - 2 = 65534$ (usable)
 Number of subnets :-
 borrowed bits = $16 - 8 = 8 \Rightarrow 2^8 = 256$ subnets.

Subnet address : 8.1.0.0
 1st valid host : 8.1.0.1
 Last valid host : 8.1.255.254
 Broadcast Address : 8.1.255.255

cii) 130.4.102.1/24
 Class : B (128-191)
 Default mask : 255.255.0.0
 Given mask : 255.255.255.0 (24)
 Network bits : 24, Host bits = 8
 Hosts per subnet = $2^8 - 2 = 254$ (usable)
 Number of subnets borrowed :
 $24 - 16 = 8 \Rightarrow 2^8 = 256$

Network Address : 130.4.102.0

1st valid host : 130.4.102.1

last valid host : 130.4.102.254

Broadcast : 130.4.102.255

(iii) 130.4.102.1/22

Class : B

Default mask : 255.255.0.0

Given mask (122) : 255.255.252.0

Network bits : 22, Host bits = 10

Hosts per subnet : $2^{10} - 2 = 1022$ (usable)

Number of subnets : $22 - 16 = 6 \Rightarrow 2^6 = 64$

1st valid : 130.4.100.1

last valid : 130.4.103.254

Broadcast : 130.4.103.255

(iv) 199.1.1.100/27

Class : C

Default mask : 255.255.255.0

Given mask : 255.255.255.224

Network bits : 27, Host bits : 5

Hosts per subnet : $2^5 - 2 = 30$ (usable)

Number of subnets : $27 - 24 = 3 \Rightarrow 2^3 = 8$ subnets

1st valid : 199.1.1.97

last valid : 199.1.1.126

Broadcast : 199.1.1.127

2. A host in a class C network has been assigned an IP Address 192.168.17.9. Find the number of addresses in the block, the first address, and the last address.

Default mask : 255.255.255.0
 Number of Addresses in block = 2^8
 256 (Total) (usable = 254)
 First Address :- 192.168.17.0
 Last Address :- 192.168.17.254

3. An address in block is given as 185.²⁸~~168~~.17.9. Find the number of addresses in the block, the first address and the last address.

Number of Addresses in block = 2^{16}
 65536 (Total) (usable = 65534)
 First Address :- 185.28.0.0
 Last Address :- 185.28.255.255

4. A block of address is granted to small organization, we know that one of the address is 205.16.37.39/28. What is the first address, last address, number of addresses in block?

First Address :- 205.16.37.33

Last Address :- 205.16.37.46

Number of addresses :- $2^{(32-28)} = 2^4$
 $= 16$ (Total) (14 usable)

5. Subnet the IP address 216.21.5.0 into 30 hosts in each subnet. Find class, Default mask

Class :- C, Default :- 124

$2^5 = 32$, so host bit required = 5.

new mask = $32 - 5 = 27$.

bit borrowed = $8 - 5 = 3$.

Number of subnets = $2^3 = 8$.

8 subnets, block size = 32.

1. 216.21.5.0/27 - 216.21.5.31

2. 216.21.5.32 - 216.21.5.63

3. 216.21.5.64 - 216.21.5.95

4. 216.21.5.96 - 216.21.5.127

5. 216.21.5.128 - 216.21.5.159

6. 216.21.5.160 - 216.21.5.191

7. 216.21.5.192 - 216.21.5.223

8. 216.21.5.224 - 216.21.5.255

6. Subnet the IP address 192.10.20.0 into 52 hosts in each subnet. Find class, default mask, Bit Borrowed, new subnet mask, No. of Hosts and subnet, Network ranges.

Class C (Default: 124)

Need host more than 52, $2^6 = 64$
 host bit required = 6, new mask = $32 - 6 = 26$
 bit borrowed = $8 - 6 = 2$.

Number of subnets = $2^2 = 4$.

host per subnet = $2^6 - 2 = 62$.

1. 192.10.20.0/26 - 192.10.20.63

2. 192.10.20.64 - 192.10.20.127

3. 192.10.20.128 - 192.10.20.191

4. 192.10.20.192 - 192.10.20.255

7. a) Device A: 172.16.17.30/20

Here we have to find the subnet mask for device A.

Mask /20 $\Rightarrow$ 255.255.240.0

240 $\Rightarrow$ 4 bits third octet $\Rightarrow$ 11110000 = 240

block size in 3rd octet = $256 - 240 = 16$

Network address : 172.16.16.0/20

First valid : 172.16.16.1

last valid : 172.16.31.254

Broadcast address : 172.16.31.255

b) Device B: 172.16.28.15/20

mask 120 (=) 255.255.240.0

third octet = 28

block size = 255 - 240 = 16.

network address : 172.16.16.0/20.

First valid : 172.16.31.1

last valid : 172.16.31.254

Broadcast : 172.16.31.255